APEX FX TRADING BOT

Product Requirements Document

Volume II — Advanced Systems & Professional Grade Enhancements

Document Owner	Product & Engineering Team
Version	2.0 — Institutional Enhancement Layer
Complements	APEX FX PRD Volume I — Version 1.0, April 2026
Date	April 2026
Classification	**CONFIDENTIAL — Internal Use Only**

This document is the direct continuation and professional enhancement of Volume I. It does not repeat foundational content. It introduces the eight critical institutional-grade systems that separate a well-designed retail bot from a system capable of surviving and improving across multiple market cycles.

11. Execution Quality & Fill Analysis Engine

Volume I specifies that slippage must be modeled in backtesting. Volume II mandates a live execution quality tracking system — the FillAnalyzer module — that runs on every completed trade and benchmarks actual execution against expected execution. Without this, the bot cannot distinguish between a strategy losing its edge and a broker degrading order quality.

11.1 FillAnalyzer Module

Every order routed through the execution engine must pass through a post-fill analysis cycle before the trade record is closed. The following metrics are captured per fill:

Metric	Definition	Alert Threshold
Execution Shortfall	Expected price minus actual fill price in pips	> 0.5 pips avg over 20 trades
Spread at Submission	Bid/ask spread recorded at order submission time	Logged; no direct threshold
Spread at Fill	Bid/ask spread at the moment of confirmed fill	Spread expansion > 30% vs submission
Fill Latency (ms)	Time elapsed from order send to broker confirmation	> 350ms average triggers VPS review
Requote Rate	Percentage of orders rejected and requoted by broker	> 3% over rolling 50 orders
Partial Fill Rate	Orders filled at less than the requested lot size	> 5% triggers lot size review

11.2 Spread Percentile Filter

The current Volume I threshold (e.g. block if spread > 2.0 pips) is static and time-blind. A 1.8 pip spread at 03:00 UTC is abnormal; at 08:30 UTC London open it is routine. Replace static thresholds with a dynamic spread percentile model:

- For each instrument, maintain a rolling 20-day distribution of spreads, segmented into 30-minute session windows (00:00–00:30, 00:30–01:00, etc.)
- At trade time, calculate the current spread's percentile within the appropriate session window.
- Block entry if current spread exceeds the 80th percentile for that window on that instrument.
- Log the spread percentile for every trade. Include in weekly performance reports.

Implementation note: The session spread distributions must be rebuilt every Monday using the prior week's tick data. Do not use static reference tables — markets evolve seasonally and across rate cycles.

11.3 Broker Quality Scoring

The FillAnalyzer aggregates weekly into a Broker Quality Score (BQS) — a composite 0–100 rating reflecting execution health. If the BQS falls below 65 for two consecutive weeks, the bot sends a critical Telegram alert recommending broker review. The BQS is computed as:

Component	Weight	Scoring
Average execution shortfall vs backtest assumption	35%	0 pip = 100 pts
Requote rate over rolling 50 orders	30%	0% = 100 pts
Average fill latency	20%	< 100ms = 100 pts
Spread expansion at fill vs submission	15%	0% expansion = 100 pts

12. Kelly Criterion & Adaptive Position Sizing

Volume I uses fixed risk percentages per instrument (e.g. 1.5% for EUR/USD). This is a sound starting point but does not account for the changing statistical edge of each strategy module over time. Volume II replaces fixed risk with a fractional Kelly sizing model that dynamically adjusts per-trade risk based on rolling live performance.

12.1 Kelly Formula Implementation

The Kelly Criterion defines the optimal fraction of capital to risk on a bet with known edge:

$$f^* = (W \times R - (1 - W)) \div R$$

Where: W = rolling win rate | R = average win ÷ average loss | f* = Kelly fraction

APEX FX applies Fractional Kelly at 25% of the full Kelly value to prevent over-betting and account for estimation error in live win rate calculations. The final risk percentage per trade is:

```
Effective Risk % = max(Min Floor, min(Max Cap, f* × 0.25))
```

12.2 Per-Instrument Kelly Parameters

Pair	Lookback Window	Min Risk Floor	Max Risk Cap	Recalc Frequency	Activation
EUR/USD	Last 50 trades	0.5%	2.0%	Weekly (Friday close)	≥ 30 live trades
GBP/USD	Last 50 trades	0.4%	1.8%	Weekly (Friday close)	≥ 30 live trades
USD/JPY	Last 50 trades	0.5%	2.0%	Weekly (Friday close)	≥ 30 live trades
USD/CHF	Last 40 trades	0.3%	1.5%	Weekly (Friday close)	≥ 25 live trades
USD/CAD	Last 50 trades	0.4%	1.8%	Weekly (Friday close)	≥ 30 live trades
XAU/USD	Last 30 trades	0.25%	1.0%	Weekly (Friday close)	≥ 20 live trades

Until the activation threshold (minimum live trades) is reached, the bot uses the fixed risk percentages defined in Volume I Section 5.3. Kelly sizing is the graduate regime — it requires real performance data to function correctly and must never be applied to insufficient samples.

12.3 Cold Streak Risk Ladder

After any sequence of three or more consecutive losses on a single instrument, the following ladder is applied regardless of Kelly calculation:

Consecutive Losses	Risk Multiplier	Trades Before Restore	Restore Condition
3 consecutive losses	**× 0.75**	Next 5 trades	2 wins in next 5
5 consecutive losses	**× 0.50**	Next 8 trades	3 wins in next 8
7+ consecutive losses	**PAIR PAUSED**	48 hours minimum	Manual review required

13. Live Regime Drift Detector & Re-Optimization Trigger

Volume I's RDE classifies market conditions on each bar. However, it does not detect when a strategy's edge is decaying due to broader structural market regime changes — a multi-week or multi-month shift in how a pair behaves. This section specifies the Regime Drift Detector (RDD), which operates at the strategy level rather than the bar level.

13.1 RDD Architecture

The RDD is a separate scheduled process that runs every Friday at 21:00 UTC after market close. It evaluates the last 30 completed trades for each instrument strategy module and compares rolling live performance against the backtest performance baseline established during pre-deployment validation.

RDD Status	Trigger Condition	Automated Response
GREEN — Healthy	Live profit factor ≥ 70% of backtest baseline	No action. Full sizing active.
AMBER — Warning	Live profit factor 50–70% of backtest baseline for 1 week	Position sizes reduced to 60%. Telegram alert sent.
RED — Degraded	Live profit factor < 50% of baseline for 2 consecutive weeks	Pair suspended. Re-optimization pipeline triggered.
CRITICAL — Override	Any single week where realized loss > 2× worst backtest week	Pair suspended immediately. Manual review required before reinstatement.

13.2 Re-Optimization Pipeline

When the RDD flags a pair as RED, the following automated pipeline is initiated:

- Step 1 — Parameter Snapshot: Current live parameters are exported and timestamped to the version control repository.
- Step 2 — Data Refresh: The last 12 months of tick-level data for the affected instrument is pulled from the historical data provider (Dukascopy preferred).
- Step 3 — Walk-Forward Re-Run: The strategy is re-optimized on the refreshed dataset using identical walk-forward methodology from Volume I Section 7.1 (70% in-sample, 30% out-of-sample, rolling).
- Step 4 — Acceptance Gate: New parameters must clear all minimum backtest acceptance criteria from Volume I Section 7.2 before promotion.
- Step 5 — Paper Trading Validation: Re-optimized strategy runs on demo account for minimum 2 weeks. Performance must not deviate more than 20% from new backtest expectations.
- Step 6 — Staged Re-Entry: Strategy returns to live at 50% position sizing for 3 weeks before returning to full Kelly-adjusted sizing.

Re-optimization must never be performed on live parameters mid-week. All parameter changes take effect at Sunday 22:00 UTC, after the weekly RDD run and before Monday's Asian session opens.

13.3 Strategy Version Control

Every parameter set used in live trading must be committed to the project's private Git repository with a standardized tag format: STRATEGY_PAIR_LIVE_YYYYMMDD. This enables instant rollback if a parameter update causes unexpected behavior in live trading. The bot reads strategy parameters from YAML config files at startup — never from hardcoded values — making version control seamless.

14. Multi-Factor Signal Scoring System

Volume I uses a binary gate model: if all entry conditions are true, a trade fires. This approach treats all confirmations as equally weighted pass/fail gates and produces no differentiation between a marginally valid signal and an exceptionally high-confidence one. Volume II replaces this with a weighted confluence scoring system — SignalScorer — that rates every potential entry on a 0–100 scale.

14.1 SignalScorer Architecture

Each strategy module submits a signal candidate to the SignalScorer. The scorer evaluates five factor groups, each with a defined maximum contribution. Only signals scoring above the minimum threshold proceed to the risk engine.

Factor Group	Max Score	Scoring Logic
Trend Alignment	30 pts	D1 + H4 + H1 all aligned = 30. Two aligned = 20. One = 10. None = 0.
Momentum Confirmation	25 pts	RSI in ideal zone + MACD confirming = 25. One of two = 12. Neither = 0.
Regime Quality	20 pts	ADX > 30 (trend) or Band Squeeze extreme (breakout) = 20. ADX 25–30 = 12. Below = 0.
Session & Timing	15 pts	Optimal session window = 15. Secondary window = 8. Outside preferred window = 0.
Execution Quality	10 pts	Spread < 40th percentile for window = 10. 40–70th = 6. 70–80th = 3. Above = 0.

14.2 Score Thresholds & Position Sizing Modifiers

Score Range	Signal Grade	Action	Position Size Modifier
0 – 59	**REJECT**	No trade opened	—
60 – 74	**MARGINAL**	Trade opened at reduced size	**× 0.60**
75 – 84	**STANDARD**	Trade opened at full Kelly size	**× 1.00**
85 – 100	**HIGH CONVICTION**	Trade opened at enhanced size	**× 1.25**

The × 1.25 High Conviction multiplier is subject to all existing caps from Volume I Section 5.3. It cannot cause a position to exceed the maximum lot size or breach portfolio heat limits defined in Section 16. Sizing always resolves to the most conservative applicable constraint.

14.3 Signal Score Logging & Backcorrelation

Every signal score is stored in the structured trade log alongside the outcome. The PerformanceAnalyzer (Section 17) runs a weekly backcorrelation to verify that higher scores genuinely produce better outcomes. If the score-to-outcome correlation drops below 0.4 for any instrument over 50 trades, the scorer weights for that instrument are flagged for recalibration.

15. Volatility Regime Scaling Layer

Volume I uses ATR-based stop distances that adapt to local price volatility. However, no adjustment is made to position size based on the broader macro-volatility environment. A system trading at full size during a post-FOMC shock week faces fundamentally different risk than the same system in a calm trending week. The Volatility Regime Scaler (VRS) is a portfolio-level multiplier applied before the Kelly position size is finalized.

15.1 VRS Calculation

The VRS computes a scalar value between 0.40 and 1.00 that multiplies every position size across all instruments simultaneously. The scalar is recalculated on every H4 bar close using two inputs:

- Realized Volatility Ratio: Current 5-day realized vol (sum of daily ATR) divided by the 60-day average realized vol for that instrument. A ratio above 1.4 indicates an elevated volatility environment.
- Cross-Instrument Correlation Spike: If three or more instruments simultaneously show ATR expansion above 1.3× their 20-day average, a correlation spike is declared — a signal that macro risk is affecting the entire portfolio simultaneously.

Condition	Realized Vol Ratio	VRS Scalar	Portfolio Effect
Calm — Low vol	< 0.8×	**1.00**	Full sizing active
Normal	0.8× – 1.2×	**1.00**	No adjustment
Elevated vol	1.2× – 1.5×	**0.70**	All positions at 70% of target size
High vol	1.5× – 2.0×	**0.50**	All positions at 50% of target size
Crisis — Correlation spike	> 2.0× or 3+ pairs spiking	**0.40**	Minimum sizing. No new positions on affected pairs.

15.2 XAU/USD Special Volatility Rules

Gold requires separate volatility handling because its ATR can expand by 300–500% during geopolitical or macro shock events. In addition to the standard VRS scalar:

- If XAU/USD ATR(14) on H4 exceeds 2.5× its 20-day ATR average, XAU/USD is placed in Defensive Mode: no new long entries. Only short scalps permitted on extreme intraday extensions above the H4 200 EMA, and only after the VRS scalar drops back below 1.8×.
- XAU/USD Defensive Mode has a minimum 6-hour hold time. It cannot be exited in less than 6 hours even if the ATR ratio normalizes faster.
- During Defensive Mode, all existing XAU/USD open positions have their trailing stop distance reduced by 30% — protecting gains as the outsized move extends.

16. Portfolio Heat Monitoring & Drawdown Sequencing

Volume I establishes per-pair daily loss limits and a global 8% daily circuit breaker but does not monitor the aggregate open risk exposure of simultaneous positions. This gap creates scenarios where four positions are open simultaneously, each at their individual risk limit, producing a total portfolio heat of 5–6% — a single correlated macro shock away from triggering the global circuit breaker. The Portfolio Heat Monitor (PHM) closes this gap.

16.1 Portfolio Heat Definition & Enforcement

Portfolio Heat is defined as the sum of current potential loss across all open positions if every stop-loss were hit simultaneously at the current stop price. It is expressed as a percentage of current account equity and recalculated on every tick for open positions.

Portfolio Heat Level	Threshold	Rule
Cold — low risk	< 2.0%	All pairs may open new positions freely.
Warm — moderate risk	2.0% – 3.0%	New positions only permitted if signal score ≥ 80 (High Conviction).
Hot — elevated risk	3.0% – 4.0%	No new positions across all instruments. Existing positions managed normally.
Critical — action required	> 4.0%	No new positions. The lowest-conviction open trade (by signal score) is reduced by 50% via partial close. Telegram alert sent.

16.2 Drawdown Sequencing Rules

Loss does not occur in a vacuum — consecutive losses create psychological and statistical pressure that static systems ignore. The following sequencing rules govern how the bot behaves during a drawdown phase, layered on top of the Kelly cold streak ladder from Section 12.3:

- After any 3% account drawdown from the rolling weekly high, the bot enters Cautious Mode: all new position sizes are capped at 75% of Kelly-calculated size until the account recovers 1.5% from the drawdown trough.
- After any 5% account drawdown from the rolling monthly high, the bot enters Recovery Mode: new positions capped at 50% of Kelly size. XAU/USD new positions are blocked entirely. Mode exits when account recovers 2% from trough.
- After any 8% drawdown from the rolling monthly high (matching the global circuit breaker threshold from Volume I), all trading halts. A daily performance report is generated and sent automatically. Manual review is required before restart.
- Drawdown levels are measured from the highest rolling equity watermark, not from the initial account balance. This ensures the system protects earned profits with the same rigor it protects initial capital.

16.3 Maximum Simultaneous Open Trades

In addition to heat monitoring, the following hard caps govern concurrent open positions across the entire portfolio:

Rule	Limit
Maximum simultaneous open trades (all instruments)	**4 trades maximum**
Maximum open trades in the same directional bias	**3 trades (e.g. max 3 net-long USD)**
Maximum open commodity-linked trades simultaneously (USD/CAD + XAU/USD)	**1 trade each; not both at maximum size concurrently**
Maximum trades open during news blackout window	**0 new trades; existing positions held**

17. Self-Learning Feedback Loop & Performance Analytics

A trading system that logs trades but never learns from them is fundamentally static. Volume II mandates a weekly PerformanceAnalyzer that closes the feedback loop between live trade outcomes and future strategy behavior — not by rewriting the strategy code, but by updating the contextual filters and scoring weights that govern when and how aggressively each strategy fires.

17.1 PerformanceAnalyzer Schedule & Data Inputs

- Runs every Friday at 21:30 UTC after market close and after the RDD run (Section 13).
- Ingests all completed trades from the structured JSON trade log for the prior 30 days.
- Minimum dataset: 15 completed trades per instrument for results to be actionable. Below this threshold, the run is logged but no parameter changes are made.
- All outputs are written to weekly_review_YYYYMMDD.json and committed to the Git repository.

17.2 Analysis Dimensions

The PerformanceAnalyzer clusters winning and losing trades across the following dimensions for each instrument:

Dimension	Buckets Analyzed	Actionable Output
Session Timing	Per 2-hour UTC window across the full 24h cycle	Session gate window narrowed or widened for next week
Volatility Regime at Entry	VRS scalar level at time of trade entry	VRS thresholds tightened if losses cluster at elevated vol
ADX Level at Entry	Bucketed: < 20, 20–25, 25–30, 30–40, > 40	ADX minimum for entry raised if losses cluster at low ADX
Signal Score	Score decile bands (60–69, 70–79, 80–89, 90–100)	Minimum score threshold raised if 60–74 band underperforms
Day of Week	Monday through Friday independently	Underperforming days flagged for session gate adjustment
Trade Duration	< 4h, 4–12h, 12–24h, 24–48h, > 48h	Maximum hold time adjusted per losing duration band

17.3 Parameter Change Governance

No parameter changes produced by the PerformanceAnalyzer are applied automatically to live trading without passing through a governance gate. The following rules apply:

- Maximum change per weekly cycle: any single parameter may shift by no more than 15% from its current value. Larger changes require manual approval.

- Only one parameter category may be updated per instrument per week. If session gate and ADX minimum both show issues, priority is given to the higher-impact one.
- All changes are applied at Sunday 22:00 UTC (same window as re-optimization from Section 13). No mid-week parameter changes permitted.
- Every change is logged with the statistical justification (sample size, win rate delta, confidence level). Undocumented changes are rejected by the config loader.

The self-learning loop is designed to make micro-corrections — tightening a session window by 30 minutes, raising the minimum ADX from 25 to 27 — not wholesale strategy rewrites. Wholesale rewrites go through the full re-optimization pipeline in Section 13.

18. COT Report Integration & Sentiment Overlay

The Commitment of Traders (COT) report, published weekly by the CFTC every Friday at 15:30 UTC, discloses the net positioning of three categories of market participants for every futures contract: Commercial hedgers, Non-commercial speculators (large funds), and Non-reportable (retail). This is the only free, official dataset showing institutional positioning in real time — and it is systematically ignored by most retail bots.

18.1 COT Data Pipeline

- Source: CFTC Disaggregated COT report (https://www.cftc.gov/dea/newcot/deacot.htm). Use the CSV feed for automated parsing.
- Relevant contracts: EUR/USD futures (Euro FX), GBP/USD (British Pound), USD/JPY (Japanese Yen), USD/CHF (Swiss Franc), USD/CAD (Canadian Dollar), Gold (COMEX — directly applicable to XAU/USD).
- Update schedule: every Friday after 16:00 UTC. Data reflects positions as of the prior Tuesday close. Factor the Tuesday-to-Friday lag into analysis — COT is a sentiment gauge, not a precise timing tool.
- Parser implementation: a lightweight Python script (cot_parser.py) downloads the CSV, filters for the six relevant contracts, and writes normalized data to cot_data.json. The signal engine reads this file on strategy initialization each week.

18.2 COT Signal Construction

Raw COT numbers are not meaningful in isolation. The actionable signal is the net speculative position expressed as a percentile of its own 3-year range — the COT Index:

```
COT Index = (Current Net Spec − 3Y Min) ÷ (3Y Max − 3Y Min) × 100
```

0 = most net short in 3 years | 100 = most net long in 3 years

18.3 COT Filter Rules by Instrument

Pair	Long Signal Confirmation	Short Signal Confirmation	Contraindication — Fade Signal
EUR/USD	COT Index > 55 (specs net long, confirm uptrend)	COT Index < 45 (specs net short, confirm downtrend)	COT Index ≥ 90 (extreme long — mean reversion risk)
GBP/USD	COT Index > 60	COT Index < 40	COT Index ≥ 90 or ≤ 10
USD/JPY	JPY COT Index < 40 (specs short JPY = long USD/JPY)	JPY COT Index > 60	JPY COT ≤ 10 (extreme short JPY, reversal risk)

XAU/USD	Gold COT Index > 50 and rising week-over-week	Gold COT Index < 40 and falling	Gold COT Index ≥ 95 (max spec long; historically marks tops)

COT confirmation adds 10 bonus points to the SignalScorer (Section 14) when the COT direction aligns with the trade direction. COT contraindication deducts 15 points — potentially converting a STANDARD signal to MARGINAL or a MARGINAL signal to REJECT.

19. Time-Based Exit Rules & Trade Duration Management

Volume I defines stop-loss, take-profit, and trailing stop logic but contains no rule governing how long a trade may remain open. An open trade that has neither hit its target nor its stop after an extended period represents dead capital — frozen equity that could be deployed elsewhere and a growing overnight risk exposure. Time-based exits are mandatory in professional systematic trading.

19.1 Maximum Hold Time by Strategy Type

Pair	Strategy Type	Max Hold Time	Time-Based Exit Rule
EUR/USD	Trend Following	**36 hours**	Exit at next H1 close if no forward progress after 12 hours
GBP/USD	Momentum Breakout	**18 hours**	Momentum trades must resolve quickly. Exit at next H1 close after 8 hours with < 0.5× ATR profit.
USD/JPY	Carry Momentum	**48 hours**	Carry trades can trend for multiple sessions. Exit if ADX drops below 20 on H4.
USD/CHF	Mean Reversion	**12 hours**	Mean reversion must succeed within the session. Exit at session close regardless of P&L.
USD/CAD	Oil-Correlated Trend	**36 hours**	Exit immediately if WTI crude reverses the directional signal that triggered entry.
XAU/USD	Trend + Breakout	**72 hours**	Gold macro trends are long-duration. However, exit if H4 200 EMA is breached in the opposite direction.

19.2 Forward Progress Rule

Independently of maximum hold time, any open trade that shows no forward progress — defined as price has not moved more than 0.3× ATR in the direction of the trade from its entry price — for a sustained period triggers an early exit:

- After 6 hours with no forward progress on an H1 strategy: move stop-loss to breakeven (entry price ± 0.5 pip for spread). Eliminates the loss risk; trade can only now break even or profit.
- After 10 hours with no forward progress: exit at market on the next H1 close. Frees capital for the next signal.
- Exception: XAU/USD trend trades are granted 12 and 18 hours respectively for the same thresholds, reflecting gold's slower, choppier trend development.

19.3 Weekend Position Policy

No new positions may be opened after 16:00 UTC on Friday on any instrument. All open positions at Friday 20:00 UTC are evaluated:

- If a position is in profit of at least 1.0× ATR: trailing stop is tightened to 0.5× ATR and the position is held through the weekend, accepting gap risk.
- If a position is in profit of less than 1.0× ATR: position is closed at market before 21:00 UTC Friday.
- If a position is in loss: position is always closed before 21:00 UTC Friday. No losing positions are held through the weekend.
- XAU/USD exception: due to active Asian Sunday session, gold positions may be held if the trailing stop has been activated and profit exceeds 1.5× ATR.

20. Day-of-Week Calibration & Seasonal Filters

Not all trading days are created equal. Monday suffers from gap risk, thin opening liquidity, and the unwinding of weekend news. Friday afternoon sees institutional position squaring, unpredictable late moves, and the risk of holding over the weekend. These patterns are consistent across years and pairs — yet Volume I applies identical rules to all five trading days. This section formalizes day-of-week differentiation.

20.1 Day-of-Week Trading Rules

Day	Restriction	Rationale	Exception
Monday	No new entries before 10:00 UTC. Position sizes at 70% of Kelly until 12:00 UTC.	Weekend gap risk; thin Asian→Europe handoff; positioning overhang from Friday close.	If signal score ≥ 90 and a major macro event drove a clear directional gap, full sizing permitted.
Tuesday – Thursday	Full operation. All rules apply normally.	Peak institutional activity. Highest quality order flow.	None — standard operation.
Friday	No new entries after 16:00 UTC. Position sizes at 75% after 14:00 UTC.	Institutional position squaring creates erratic, low-conviction moves in the final 2 hours.	NFP Friday: standard sizing permitted until NFP release. Blackout applies per Section 5.5.

20.2 Monthly Calendar Filters

The following recurring monthly events trigger automatic adjustments regardless of day of week:

- Month-end rebalancing (last 2 trading days of each month): Position sizes reduced to 60% across all pairs. Month-end USD fixing flows (particularly at 16:00 UTC) create artificial directional spikes that reverse within hours.
- Quarter-end (last 5 trading days of March, June, September, December): Same as month-end, extended to 5 days. This is the highest institutional rebalancing period of the year. Trend signals should be treated with additional skepticism — confirm with COT and spread percentile before entry.
- NFP week (first Friday of each month): Starting the Wednesday before NFP, XAU/USD and EUR/USD operate at 75% position sizing. The day before NFP (Thursday), all pairs operate at 60%.
- FOMC meeting weeks (8 times per year): XAU/USD and USD-paired instruments operate at 65% sizing from Tuesday through the Wednesday release. Post-release, the gold session breakout strategy (Volume I Section 4.7) takes priority — trade the breakout 15 minutes after the statement, not before.

20.3 Seasonal Performance Tracking

The PerformanceAnalyzer (Section 17) maintains a rolling 12-month win rate and profit factor breakdown by calendar month for each instrument. August historically shows reduced liquidity across

all pairs due to European summer vacations — institutionally thin markets. If August win rates fall below 45% for any instrument in the prior year's data, that instrument's minimum signal score is raised from 75 to 82 for the entire month of August in the current year.

21. Additional Professional-Grade Filters

This section documents five additional filters and rules that individually have modest impact but collectively close the final gap between a well-built retail bot and a system operating at institutional quality. Each rule is the result of commonly overlooked edge cases that cause unnecessary losses in production systems.

21.1 Spread-to-ATR Ratio Filter

Before any trade entry, regardless of strategy, the following check must pass:

```
BLOCK ENTRY if: Current Spread ÷ ATR(14) on H1 > 0.15
```

Meaning: spread must not exceed 15% of the expected daily range

This single filter eliminates entries during hidden liquidity crises — periods where the spread is high relative to the available move. A EUR/USD trade entered with a 1.5 pip spread when the H1 ATR is only 8 pips means 18.75% of the expected range is consumed in transaction costs before price moves a single pip in your favor.

21.2 Broker Redundancy & Secondary Route

Volume I specifies a primary Python bot with a backup MT5 EA for drawdown protection. Volume II extends this with a secondary broker account specification:

- A secondary ECN broker account must be funded with a minimum of 20% of primary account capital.
- The backup MT5 EA (MQL5) is deployed on the secondary account. It monitors the primary account's equity via a shared signal file. If the primary account's equity drops more than 2% in any 15-minute window AND the Python bot heartbeat is absent, the backup EA opens offsetting positions on the secondary account to hedge exposure.
- The secondary account is never used for independent trading — only for emergency hedging. It is reviewed and reconciled weekly.

21.3 Requote & Retry Logic (Enhanced)

Volume I mentions retry logic with exponential backoff. Volume II specifies the exact retry parameters:

Attempt	Wait Before Retry	Slippage Tolerance	Action If Still Fails
1st attempt	Immediate	0.3 pips	Proceed to attempt 2
2nd attempt	500ms	0.5 pips	Proceed to attempt 3
3rd attempt	1,500ms	0.8 pips	Proceed to attempt 4

4th attempt (final)	4,000ms	1.2 pips	Signal abandoned. Log as MISSED_SIGNAL. Alert sent.

21.4 Rollover / Swap Cost Filter

All positions held past 22:00 UTC incur overnight swap costs (rollover). For short-duration strategies (USD/CHF mean reversion, GBP/USD momentum breakout), rollover costs can eliminate small profits entirely. Before opening any position within 2 hours of the 22:00 UTC rollover window:

- Fetch current swap rate from MT5 (symbol.swap_long / symbol.swap_short).
- Calculate: if holding until the minimum viable take-profit, does the expected profit exceed rollover cost by at least 2×? If not, delay entry until after rollover.
- Applies to: GBP/USD momentum entries after 20:00 UTC, USD/CHF mean reversion entries after 20:30 UTC.
- Does not apply to: EUR/USD, USD/JPY, or XAU/USD trend trades where the expected hold time and profit target comfortably absorb rollover.

21.5 Market Impact Awareness

At lot sizes approaching the volume caps defined in Volume I Section 5.3, APEX FX must be aware that its own orders can move the market, particularly for less liquid instruments. The following maximum single-order sizes prevent material self-impact:

Instrument	Max Single Order (lots)	If Target Size Exceeds Limit
EUR/USD	**1.0 lots per order**	Split into 2 orders 30 seconds apart, same TP/SL levels.
GBP/USD	**0.75 lots per order**	Split into 2 orders 45 seconds apart.
XAU/USD	**0.25 lots per order**	Split into 2 orders 60 seconds apart. Gold is less liquid per unit value.

22. Enhanced Monitoring, Reporting & Alerting

Volume I specifies Telegram alerts for individual trade events and a daily performance summary. Volume II upgrades the monitoring layer to match what a professional fund operations team would expect: tiered alerts, a structured weekly report, and real-time portfolio health dashboards.

22.1 Alert Tiering

Tier	Trigger Examples	Delivery	Response SLA
INFO	Trade opened, trade closed, trailing stop moved	Telegram (silent notification)	No action required
WARNING	Spread percentile > 70th, VRS scalar drops to 0.70, RDD status → AMBER, Kelly recalculated with unusual shift	Telegram (standard notification)	Review within 4 hours
CRITICAL	Daily circuit breaker triggered, RDD → RED, BQS < 65, pair suspended, bot heartbeat missed	Telegram (with sound) + Email	Review within 30 minutes
EMERGENCY	Global 8% drawdown circuit breaker, VPS connectivity lost, MT5 terminal crash, backup EA activated	Telegram (with sound) + Email + SMS (if configured)	Immediate manual intervention

22.2 Weekly Performance Report Structure

Generated every Saturday at 08:00 UTC, covering the prior Monday–Friday trading week. Delivered via Telegram document and written to weekly_report_YYYYMMDD.json. Report contains the following mandatory sections:

- Executive Summary: Net P&L, trades taken, win rate, profit factor, maximum intraweek drawdown, Sharpe contribution.
- Instrument Breakdown: Per-pair win rate, average R:R achieved, average signal score of winning vs losing trades, number of signals rejected (with reason distribution).
- Execution Quality: Average fill latency, execution shortfall, requote rate, Broker Quality Score (BQS) — all vs prior week.
- Risk Snapshot: Peak portfolio heat reached during the week, VRS scalar range (min/max), days spent in Cautious or Recovery Mode.
- RDD & Kelly Status: Current status (GREEN/AMBER/RED) per pair, current effective risk % per pair vs fixed baseline, Kelly edge estimate per pair.
- COT Update: Current COT Index per instrument vs prior week — directional shift flagged if change > 15 points.
- Next Week Outlook: High-impact events calendar for the coming week with pre-declared adjustments (e.g. NFP Thursday sizing reduction, FOMC blackout times).

23. Updated Technical Architecture & Module Registry

This section provides the complete updated module registry, incorporating all new modules introduced in Volume II alongside the original modules from Volume I Section 6.

23.1 Complete Module Registry

Module	Volume	Responsibility
mt5_feed.py	Vol I	OHLCV data provider, real-time tick subscription
calendar_feed.py	Vol I	Economic calendar, news blackout manager
commodity_feed.py	Vol I	WTI crude oil price feed for USD/CAD signal confirmation
regime_detector.py	Vol I	RDE: classifies TRENDING / RANGING / BREAKOUT / AVOID per bar
[pair]_strategy.py (×6)	Vol I	Per-instrument strategy modules, indicator computation, raw entry signals
position_sizer.py	Vol I + II	ATR-based sizing (Vol I) + Kelly fraction + VRS scalar + signal score modifier + cold streak ladder
drawdown_monitor.py	Vol I + II	Daily and rolling DD tracker + Cautious/Recovery/Halt mode logic
correlation_manager.py	Vol I + II	Cross-pair exposure control + directional bias cap + commodity pair cap
order_router.py	Vol I + II	MT5 order placement + enhanced retry logic + order splitting for market impact
spread_checker.py	Vol I + II	Static spread filter (Vol I) + spread percentile filter + spread-to-ATR filter
fill_analyzer.py	Vol II NEW	Post-fill execution quality analysis, BQS computation, slippage tracking
signal_scorer.py	Vol II NEW	Multi-factor signal scoring (0–100), threshold gating, position modifier output
kelly_sizer.py	Vol II NEW	Rolling win rate / R tracker, fractional Kelly calculation, risk floor/cap enforcement
volatility_scaler.py	Vol II NEW	VRS scalar computation, realized vol ratio, correlation spike detection, XAU Defensive Mode
portfolio_heat_monitor.py	Vol II NEW	Real-time aggregate open risk computation, heat level enforcement, partial close trigger

regime_drift_detector.py	Vol II NEW	Weekly strategy performance vs baseline comparison, RDD status, re-optimization trigger
performance_analyzer.py	Vol II NEW	Weekly self-learning analysis, dimension clustering, governed parameter update generation
cot_parser.py	Vol II NEW	CFTC COT report downloader, COT Index calculator, weekly JSON output for signal scorer
time_exit_manager.py	Vol II NEW	Max hold time enforcement, forward progress monitoring, weekend position policy
calendar_filter.py	Vol II NEW	Day-of-week rules, month-end/quarter-end filters, seasonal size adjustments
swap_filter.py	Vol II NEW	Rollover cost fetch, pre-22:00 UTC entry profitability check per instrument

23.2 Updated Pre-Trade Check Sequence

The execution loop from Volume I Section 6.4 is extended. The full pre-trade check sequence before any order is submitted is now:

- 1. Regime detection (RDE) — must not be AVOID
- 2. Session gate (per Volume I strategy specs + day-of-week filter from Section 20)
- 3. News blackout check (calendar_feed)
- 4. Spread percentile check (< 80th percentile for session window)
- 5. Spread-to-ATR ratio check (< 0.15)
- 6. Swap cost check (if within 2 hours of rollover for short-duration strategies)
- 7. COT alignment check (adds/deducts points in scorer)
- 8. SignalScorer evaluation (must achieve ≥ 60 to proceed; < 60 is hard reject)
- 9. VRS scalar application (portfolio-level volatility environment)
- 10. Kelly sizing calculation (subject to floors, caps, and cold streak ladder)
- 11. Portfolio heat check (current open risk vs heat thresholds)
- 12. Correlation exposure check (EUR+CHF, GBP+EUR, CAD+Gold caps)
- 13. Maximum simultaneous trades check (≤ 4 total open)
- 14. Market impact order splitting (if lot size exceeds single-order limits)
- 15. Order submission with server-side stop-loss and take-profit

24. Volume II Implementation Roadmap

The modules introduced in Volume II should be built in order of risk impact, not development convenience. The following phased plan integrates with the Volume I roadmap (Phases 1–4) and assumes the system is operational on live markets.

24.1 Phase 5 — Core Risk Enhancement (Weeks 1–4 post-live)

Module	Rationale for Priority
portfolio_heat_monitor.py	Highest immediate risk impact. Prevents catastrophic simultaneous drawdown across correlated positions.
volatility_scaler.py	Second highest risk impact. Prevents full-size entries during macro shock events.
spread_checker.py (update)	Add spread percentile and spread-to-ATR filters. High signal-to-effort ratio improvement.
time_exit_manager.py	Eliminates dead capital and reduces overnight gap risk. Quick to build.

24.2 Phase 6 — Signal Intelligence (Weeks 5–8 post-live)

Module	Rationale for Priority
signal_scorer.py	Replaces binary gating with weighted confluence. Requires 4 weeks of baseline data from Phase 5 to calibrate properly.
fill_analyzer.py	Builds the execution quality dataset needed for BQS and future broker evaluation.
calendar_filter.py	Day-of-week and monthly rules. Low complexity, meaningful edge improvement.
cot_parser.py	CFTC COT integration. Adds institutional positioning context to SignalScorer.

24.3 Phase 7 — Adaptive Intelligence (Weeks 9–14 post-live)

Module	Rationale for Priority
kelly_sizer.py	Requires 30–50 live trades per pair before activation. Build in Week 9, activate when trade count qualifies.
regime_drift_detector.py	Requires sufficient live performance history (minimum 8 weeks) to establish reliable baseline.
performance_analyzer.py	The self-learning loop. Requires trade history and all scoring dimensions to be operational. The final layer.

25. Revised Success Metrics — Combined Volume I + II System

With all Volume II enhancements operational, the following revised success metrics replace those in Volume I Section 2.3. These reflect the higher bar set by institutional-grade execution quality and adaptive risk management.

Metric	Vol I Target	Vol II Minimum	Vol II Stretch
Monthly Net Return	3% – 5%	**4% – 6%**	**8% – 12%**
Maximum Account Drawdown	< 15%	**< 10%**	**< 7%**
Win Rate (All Pairs)	> 50%	**> 52%**	**> 62%**
Profit Factor	> 1.4	**> 1.5**	**> 2.0**
Sharpe Ratio (Annualized)	> 1.0	**> 1.2**	**> 1.8**
Avg Execution Shortfall	Not tracked	**< 0.3 pips**	**< 0.15 pips**
Broker Quality Score (BQS)	Not tracked	**> 72**	**> 85**
Signal Score Accuracy (corr.)	Not applicable	**> 0.40**	**> 0.60**
Peak Portfolio Heat	Not tracked	**< 3.5%**	**< 2.5%**
System Uptime	> 99%	**> 99.5%**	**> 99.9%**

APEX FX TRADING BOT — Product Requirements Document Volume II

Version 2.0 • April 2026 • CONFIDENTIAL — Internal Use Only

This document must be read in conjunction with APEX FX PRD Volume I. It does not supersede Volume I — it extends it. All Volume I specifications remain in force unless explicitly overridden herein.